MPAS-URBAN QUICKSTART | GUIDE 3 OF 3

# Run the HRAIN2025 case

*Prepare the run directory, apply the 500 m mesh setting, and optionally use the HRAIN2025 NTDK cutoff modification.*

OUTCOME You will launch the three-day HRAIN2025 atmosphere experiment and understand which settings are standard, case-specific, or required only for comparison with existing historical results.

AUTHOR LIU Zhuo
AFFILIATION The Hong Kong University of Science and Technology (HKUST)

## Before you begin

Complete the recommended official MPAS tutorial and Guides 1–2. Confirm that the initial-condition file and hk_hull_500m_graph.info.part.112 exist.

Choose the executable deliberately: standard hkust-dev for normal MPAS-Urban learning and new experiments, or the verified HRAIN2025 case-specific commit 7c1610b8f41781c2433f780bd7496da63b25a920 for the optional 10 km CU cutoff and HRAIN2025 ZT compatibility setting. Soil categories are already stored in the static file: HRAIN2025 used the dataset described by Dy and Fung (2016), while Guide 2 also documents an official-BNU public approximation.

## Step 1 | Prepare the run directory

Guide 2 created an independent case under $HOME/MPAS. MPAS_BUILD is the tutorial-defined path to the compiled MPAS source tree, not an official MPAS variable. Set it again when using a new shell, select the same build used for initialization, and link all runtime files from that build into the existing case.

**SELECT THE SAME COMPILED BUILD**

```
export MPAS_ROOT="$HOME/MPAS"
export MPAS_BUILD="$MPAS_ROOT/HKUST-MPAS"

test -x "$MPAS_BUILD/atmosphere_model"
```

For the verified HRAIN2025 comparison/cutoff build, replace MPAS_BUILD with $MPAS_ROOT/HKUST-MPAS-NTDK. Do not mix files from the two builds.

For a fully comparable HRAIN2025 simulation, continue from a static file generated with the Dy and Fung (2016) dataset and the corresponding initialization setup. An official-BNU static file is a closer public approximation than default STATSGO/FAO, but it is not an exact replacement. Use separate case directories for each soil/source combination.

#### ENTER THE EXISTING CASE

```
export HRAIN_REPO="$MPAS_ROOT/MPAS-Urban-HRAIN2025"
export CASE="$MPAS_ROOT/HRAIN2025_30km_500m"

test -d "$CASE"
cd "$CASE"
test -s 30km_500m.init.nc
test -s hk_hull_500m_graph.info.part.112
```

#### COPY THE RUN CONFIGURATION

```
cp "$HRAIN_REPO/config/namelist.atmosphere" ./namelist.atmosphere
cp "$MPAS_BUILD/streams.atmosphere" ./streams.atmosphere
```

#### LINK ALL RUNTIME FILES FROM THE SELECTED BUILD

```
ln -sf "$MPAS_BUILD/atmosphere_model" ./atmosphere_model
ln -sf "$MPAS_BUILD"/default_inputs/stream_list.atmosphere.* .
ln -sf "$MPAS_BUILD"/src/core_atmosphere/physics/physics_wrf/files/*TBL .
ln -sf "$MPAS_BUILD"/src/core_atmosphere/physics/physics_wrf/files/*DATA* .
ln -sf "$MPAS_BUILD"/src/core_atmosphere/physics/physics_noahmp/parameters/*TBL .
```

#### FINAL FILE CHECK

```
ls -lh atmosphere_model
ls -lh namelist.atmosphere streams.atmosphere
ls -lh 30km_500m.init.nc hk_hull_500m_graph.info.part.112
```

BUILD CONSISTENCY NOTE The executable, streams file, stream lists, physics tables, and data links must all come from the same MPAS_BUILD. Do not relink a case after it has produced output; use a separate case directory when comparing source versions.

## Step 2 | Set the case time and mesh scale

REFERENCE NAMELIST https://github.com/Liuzh223/MPAS-Urban-HRAIN2025/blob/main/config/namelist.atmosphere

FILE TO EDIT namelist.atmosphere. Copy the repository version into the run directory and verify the following fields before launch.

CHANGE 1 - config_start_time: the first model time is 2025-08-03_00:00:00.

CHANGE 2 - config_run_duration: 3_00:00:00 covers the complete three-day case; the commented three-hour value is only a smoke test.

CHANGE 3 - config_dt: 6 seconds is the dynamics time step used by this experiment.

CHANGE 4 - config_len_disp: explicitly set 500.0 here in the running namelist. This is the first point in the tutorial where this case parameter is introduced.

#### NAMELIST.ATMOSPHERE

```
&nhyd_model
    config_start_time   = '2025-08-03_00:00:00'
    config_run_duration = '3_00:00:00'
    config_dt           = 6
    config_len_disp     = 500.0
/
```

REQUIRED CASE SETTING  Keep config_len_disp = 500.0. It explicitly records the 500 m minimum mesh spacing used by this experiment and should not be omitted even when the mesh contains nominalMinDc.

# Step 3 | Set the physics schemes and optional cutoff

FILE TO EDIT  Continue editing the &physics group in namelist.atmosphere. The following suite is the explicit HRAIN2025 case configuration; verify every value rather than relying on source-tree defaults.

## Physics suite used by HRAIN2025

The case uses WSM6 microphysics, New Tiedtke cumulus, Noah-MP land surface, YSU boundary layer and gravity-wave-drag options, cloud fraction, RRTMG longwave and shortwave radiation, the revised Monin-Obukhov surface layer, and MPAS-Urban. Use the same suite with either source tree; the standard hkust-dev build omits config_cu_disable_dx.

```
&physics
    config_microp_scheme       = 'mp_wsm6'
    config_convection_scheme   = 'cu_ntiedtke'
    config_lsm_scheme          = 'sf_noahmp'
    config_pbl_scheme          = 'bl_ysu'
    config_gwdo_scheme         = 'bl_ysu_gwdo'
    config_radt_cld_scheme     = 'cld_fraction'
    config_radt_lw_scheme      = 'rrtmg_lw'
    config_radt_sw_scheme      = 'rrtmg_sw'
    config_sfclayer_scheme     = 'sf_monin_obukhov_rev'
    config_urban_physics       = true
/
```

## Optional HRAIN2025 cutoff derivative

Use the following only when atmosphere_model was compiled from the verified HRAIN2025 case-specific commit 7c1610b8f41781c2433f780bd7496da63b25a920 on Liuzh223/HKUST-MPAS-LIU branch ntdk-grid-cutoff.

```
&physics
    config_convection_scheme = 'cu_ntiedtke'
    config_cu_disable_dx     = 10000.0   ! m; set 0.0 to disable cutoff
/
```

The value 10000.0 is a 10 km grid-spacing threshold. With config_convection_scheme = 'cu_ntiedtke', New Tiedtke convection is suppressed where local mesh spacing is below 10 km. Set config_cu_disable_dx = 0.0 to disable this cutoff; the New Tiedtke scheme then remains active according to its normal configuration. Non-positive values do not activate the threshold.

SOURCE NOTE  ntdk-grid-cutoff is a case-author derivative, not the official HKUST-MPAS main branch. The verified case commit is https://github.com/Liuzh223/HKUST-MPAS-LIU/commit/7c1610b8f41781c2433f780bd7496da63b25a920. There is no separate shell command named cutoff.

## Step 4 | Check the decomposition and streams

CHANGE 1 - namelist.atmosphere/&decomposition: use the prefix hk_hull_500m_graph.info.part. and launch 112 MPI tasks.

```
&decomposition
    config_block_decomp_file_prefix =
      'hk_hull_500m_graph.info.part.'
/
```

Use 112 MPI tasks so that the prefix resolves to hk_hull_500m_graph.info.part.112. In streams.atmosphere, point the input stream to the initial-condition file created in Guide 2 and select the desired output interval.

FILE TO EDIT streams.atmosphere. Set the input filename to 30km_500m.init.nc, choose the history filename and output interval, and verify that all relative paths are resolved from the run directory.

## Step 5 | Run and monitor

RUN

```
cd "$CASE"
export OMP_NUM_THREADS=1

mpiexec -n 112 ./atmosphere_model
```

CHECK

```
tail -n 60 log.atmosphere.0000.out
ls -lh history.*.nc 2>/dev/null || ls -lh *.nc
```

For an inexpensive first check, temporarily use a short config_run_duration, confirm that output is created, and then restore the intended three-day duration. Do not describe the short smoke test as the HRAIN2025 experiment result.

## Final note | Comparing with existing HRAIN2025 results

The HRAIN2025 historical experiment results have not yet been publicly released. This section is needed only for a future direct comparison with those existing results. It is not required for learning MPAS or for a new independent simulation.

Use the verified HRAIN2025 case-specific source commit: https://github.com/Liuzh223/HKUST-MPAS-LIU/commit/7c1610b8f41781c2433f780bd7496da63b25a920. This source contains the 10 km CU cutoff, the HRAIN2025 ZT compatibility setting, and a backported official-BNU selector for the public static-data alternative; the rainfall branch is not required.

SOIL DATA HRAIN2025 static interpolation used the updated global WRF soil map described by [Dy and Fung (2016)](). Readers needing a fully comparable HRAIN2025 simulation should consult the paper and contact its authors for the same data and processing details. An official-BNU run should be reported as a public approximation rather than an exact reproduction.

SOURCE FILE

```
src/core_atmosphere/physics/physics_wrf/module_sf_urban.F
```

Inside SFCDIF_URB, the case-specific source has two active ZT updates: one after USTAR is calculated and one inside the stability iteration. The behavior originates from upstream HKUST-MPAS commit 562bdb4e5ef63527496b6374d52def06a01a84a0: https://github.com/HKUST-MPAS/HKUST-MPAS/commit/562bdb4e5ef63527496b6374d52def06a01a84a0. Z0 is the momentum roughness length and ZT is the thermal roughness length; this setting does not redefine or recalculate Z0.

```
ZT = 0.1 * Z0
```

The verified case-specific commit already includes both active ZT = 0.1 * Z0 assignments, so no manual source edit is needed. Check out the exact commit and rebuild atmosphere_model. Do not confuse this with Z0HC_TBL = 0.1 * Z0C_TBL, which is a different calculation.

For a new independent experiment, keep the ZT implementation supplied by the selected HKUST-MPAS version and record the repository, branch, and commit. Do not modify ZT merely to complete the tutorial.

## Completion checklist

- config_start_time is 2025-08-03_00:00:00.
- config_run_duration is 3_00:00:00 for the full experiment.
- config_len_disp is explicitly 500.0.
- config_cu_disable_dx is present only for a verified cutoff-capable build; it is 10000.0 for this case, while 0.0 disables the cutoff.
- The MPI task count matches .part.112.
- For direct comparison with the original HRAIN2025 configuration, use the exact case-specific atmosphere commit and the Dy and Fung (2016) static-input dataset. The official MPAS BNU dataset is a documented public approximation, not an exact replacement.

## Primary references

- Dy and Fung (2016), updated global WRF soil map
- Official MPAS-A tutorial
- MPAS-A Quick Start Guide
- HKUST-MPAS hkust-dev
- HRAIN2025